

Sai Pushkar Sikharam

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SUMMARY

Computer Science graduate with strong experience in backend and full-stack software engineering, specializing in building scalable APIs, data-driven systems, and cloud-ready applications. Proficient in Python, FastAPI, REST APIs, React, SQL, and cloud platforms (AWS, GCP). Experienced in integrating AI services into production systems, designing clean architectures, and delivering reliable, maintainable software in academic and real-world environments.

TECHNICAL SKILLS

- **Programming Languages:** C++, Java, Python
- **Backend & API:** FastAPI, Flask, REST APIs, Authentication, Microservices
- **Frontend:** React, JavaScript, HTML, CSS
- **Databases:** SQL, MongoDB
- **Cloud:** AWS, Docker, CI/CD
- **AI & Applied ML:** Hugging Face, OpenAI API, NLP, LLM Integration

WORK EXPERIENCE

Texas A&M University - Corpus Christi

Graduate Research Assistant

Jun 2025 – May 2026

- Architected an automated GitHub CVE/CWE security auditing pipeline analysing 89 open-source repositories, correlating Issues, PRs, Commits, and Releases into unified findings.
- Implemented deterministic vulnerability detection (regex + confidence scoring), fix-version resolution (SHA-to-tag mapping), and rate-limit-aware scalable API orchestration.
- Built *Crypto Fusion*, FastAPI-based backend services exposing ML inference pipelines as production-ready APIs, integrating caching, structured logging, and robust error handling to improve reliability and reduce failed requests by 30% under high-load testing.
- Designed high-throughput data ingestion pipeline processing thousands of daily financial and new records, integrating LLM based summarization and sentiment services via asynchronous batching reducing model load by 40% & cutting inference to 5s.

Math Buddy

Python Developer

Mar 2022 – Dec 2022

- Built a Python-based desktop voice assistant using speech recognition and TTS libraries to demonstrate real-time input processing and command execution.
- Delivered Python programming instruction and designed hands-on assignments covering OOP, algorithms, and debugging practices.
- Assisted development teams in evaluating program outputs and validating the logic through systematic test case execution

PROJECTS

Talk2Campus - Fast API, ReactJS, Python, Mongo DB, OpenAI API

Aug 2025 – Dec 2025

- Built a production style AI campus assistant using FastAPI and React, implementing 10+ REST endpoints for authentication, query orchestration and structured data retrieval.
- Implemented a retrieval augmented pipeline with LLM integration, to reduce average response latency to <2s.

Crypto Currency Price Forecasting using Machine Learning-- Python, PyTorch, ML

Sept 2023 – April 2024

- Developed a machine learning pipeline for cryptocurrency price prediction using real time market data and backend API Intergration.

Helmet Detection and Number Plate Recognition using YOLOv3 in Real Time – Python

Feb 2023 – May 2023

- Built a real-time Flask-based web application capable of processing video streams at ~20 FPS.

EDUCATION

Texas A&M University – Corpus Christi

Aug 2024 – May 2026

Master's Computer Science

- **Coursework:** Machine Learning, Deep Learning, Advanced Software Engineering, DSA, Transformers, ML Ops, LLM

SCSVMV University – Kanchipuram

May 2020 – May 2024

Computer Science and Engineering

- **Coursework:** Operating Systems, Computer Networks, DBMS, OOPs, Compiler Design, Web Technologies, Cloud Computing

ACHIEVEMENTS

- **IEEE 2023:** Published research paper titled “*Helmet Detection and Number Plate Recognition using YOLOv3*”